

STEEL BEAM ESTIMATION

Current Hybrid Architecture Performance · Second to Sixth Drawing Sets

Version 10 · MODEL_VERSION 10.11.24 · Gate

P2610E3_SECOND_TO_SIXTH_FULL_POPULATION_LIVE_VISION_HYBRID_ACCURACY_BENCHMARK_AND_REPORT_V1_0 · Decision PASS_WITH_LIMITATIONS · Benchmark mode LIVE_BENCHMARK

CURRENT HYBRID ARCHITECTURE PERFORMANCE — SECOND TO SIXTH SETS

Beam identification	Bar identification	Correct of detected bars	Diameter identification
85.12%	48.17%	36.50%	77.84%

Steel / weight accuracy	Overall accuracy	HYBRID coverage	FALLBACK coverage
60.64%	57.61%	91.70%	8.30%

This report measures the current hybrid architecture on Second through Sixth drawing sets. Claude Vision is the preferred semantic interpreter after validation. The deterministic engine remains the engineering calculation authority. First Set is excluded. This is not a historical improvement study and not a production-readiness claim.

1. Executive summary

Five drawing sets were discovered dynamically and scored with the QA.2A / QA.3.0 methodology. Model beams: 530. Ground-truth beams: 605. Matched IDs: 515. Vision usable (HYBRID): 486. Deterministic fallback: 44. Pooled overall accuracy is the mean of pooled beam identification, bar identification, correct-of-detected, and kg-pooled steel/weight accuracy. Diameter is reported separately.

2. Current hybrid architecture

DXF → Beam discovery → Deterministic geometry context → Claude Vision semantic interpretation → D.1 semantic authority contract → D.2 hybrid resolution → D.3 engineering binding → D.4 shadow engineering calculation → accuracy benchmark versus estimator truth.

VISION SEMANTICS (preferred after validation): target identity, layer, physical group detection, bar count, diameter, specification, MAIN/EXTRA role, support scope, stirrup identification. DETERMINISTIC ENGINEERING: spacers, geometry, cut length, development length, anchorage, hooks/bends, stirrup geometry/zones/spacing/quantity, piece generation, weight, BBS, workbook generation. Vision-preferred does not mean blindly accepted. Ambiguous matches are withheld, not forced.

3. Benchmark scope and population

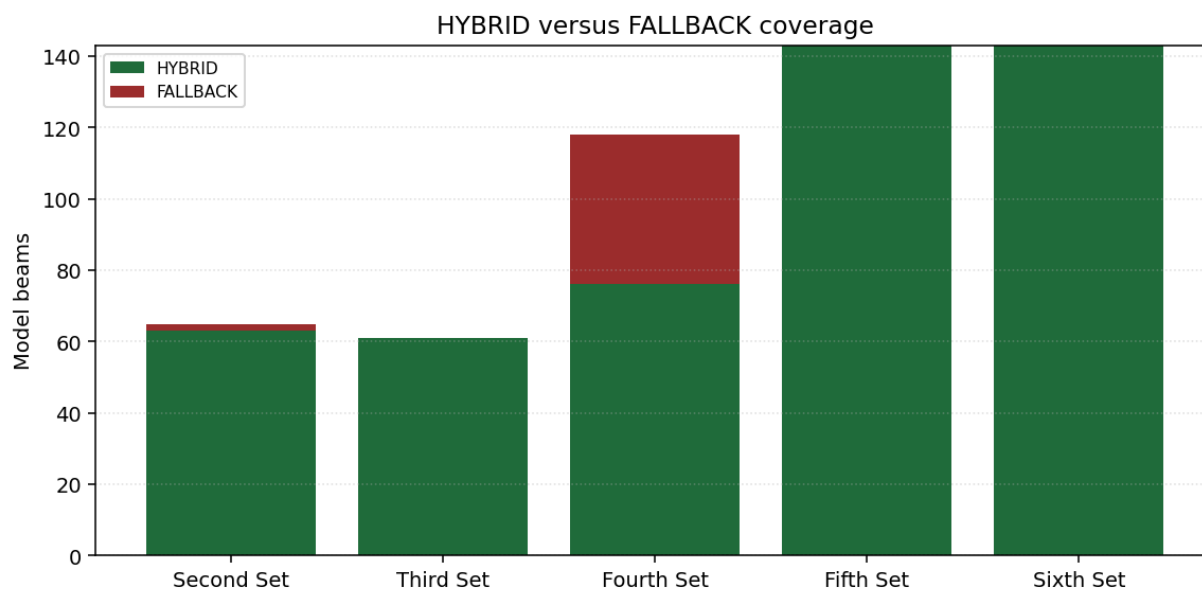
Drawing set	Model beams	GT beams	Matched	Unmatched model	Unmatched GT	Truth source
Second Set	65	67	64	1	2	ESTIMATOR_EXCEL
Third Set	61	63	57	4	4	ESTIMATOR_EXCEL
Fourth Set	118	143	112	6	22	ESTIMATOR_EXCEL
Fifth Set	143	187	143	0	42	ESTIMATOR_EXCEL
Sixth Set	143	145	139	4	6	ESTIMATOR_EXCEL

First Set is excluded from all aggregate KPIs. Unmatched beams are recorded, not silently dropped.

4. Vision execution and hybrid coverage

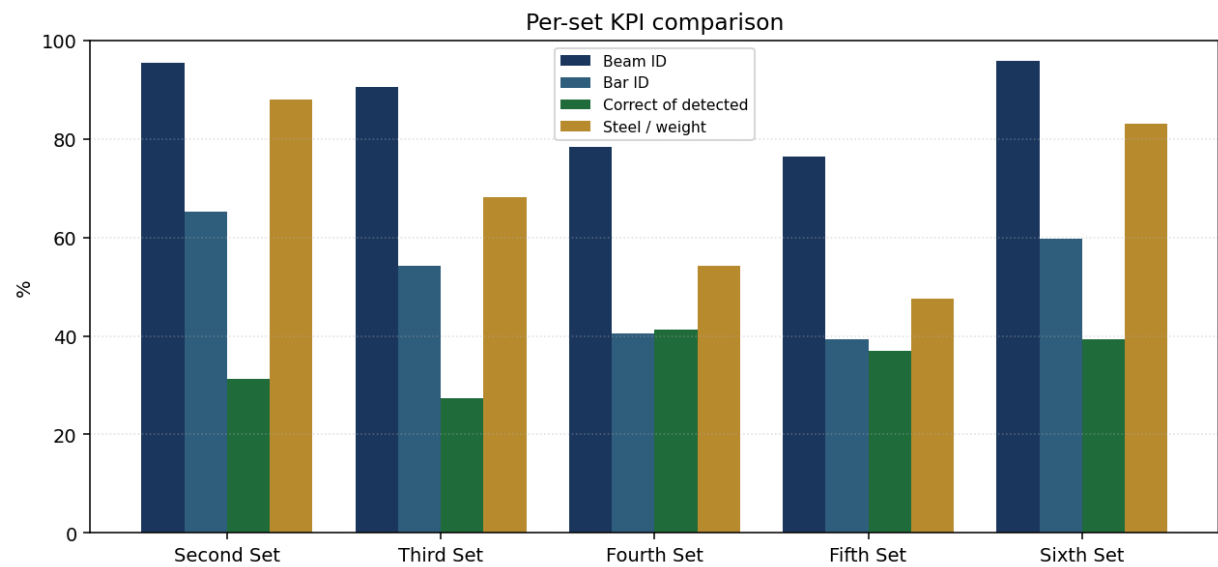
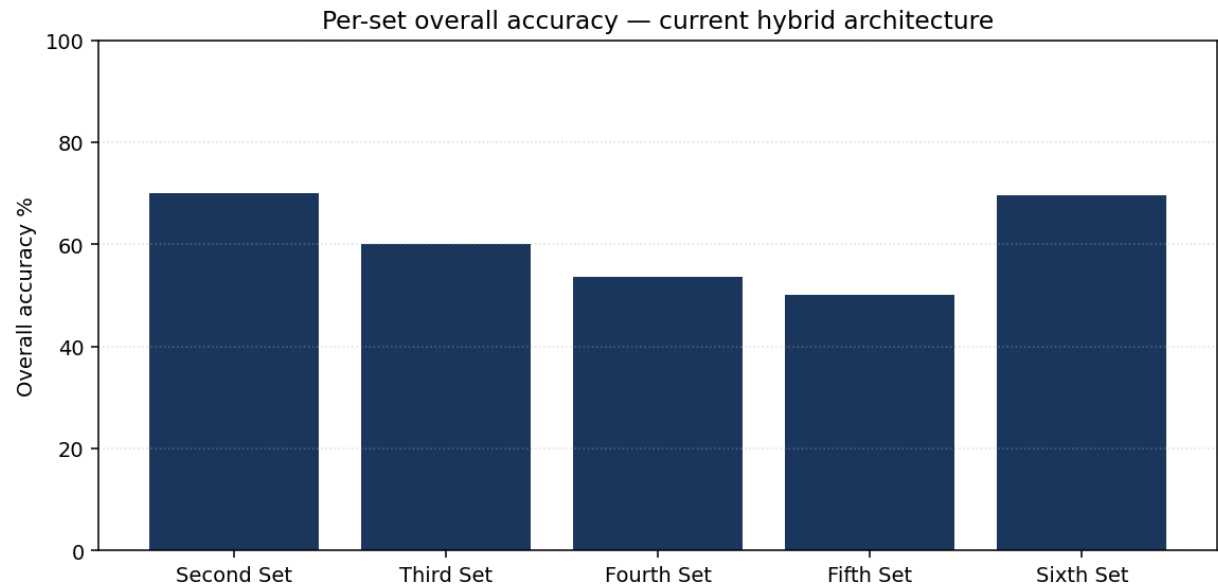
Set	Eligible	Attempted	New live	Reused	Retried	API ok	Schema	Usable	HYBRID	FALLBACK
Second Set	63	63	63	0	0	63	63	63	63	2
Third Set	61	61	61	0	0	61	61	61	61	0
Fourth Set	76	76	76	0	0	76	76	76	76	42
Fifth Set	143	0	0	143	0	143	143	143	143	0
Sixth Set	143	143	143	0	0	143	143	143	143	0

Fifth Set valid E.2 live Vision artefacts are reused when the current semantic contract, schema, source fingerprints and population match (VISION_REUSED_CURRENT_ARCHITECTURE). Second, Third, Fourth and Sixth use current live Claude (claude-sonnet-4-5) where a valid render exists. Individual Vision failures apply the D.1 fallback policy and are labelled FALLBACK, never HYBRID.



5. Accuracy by drawing set

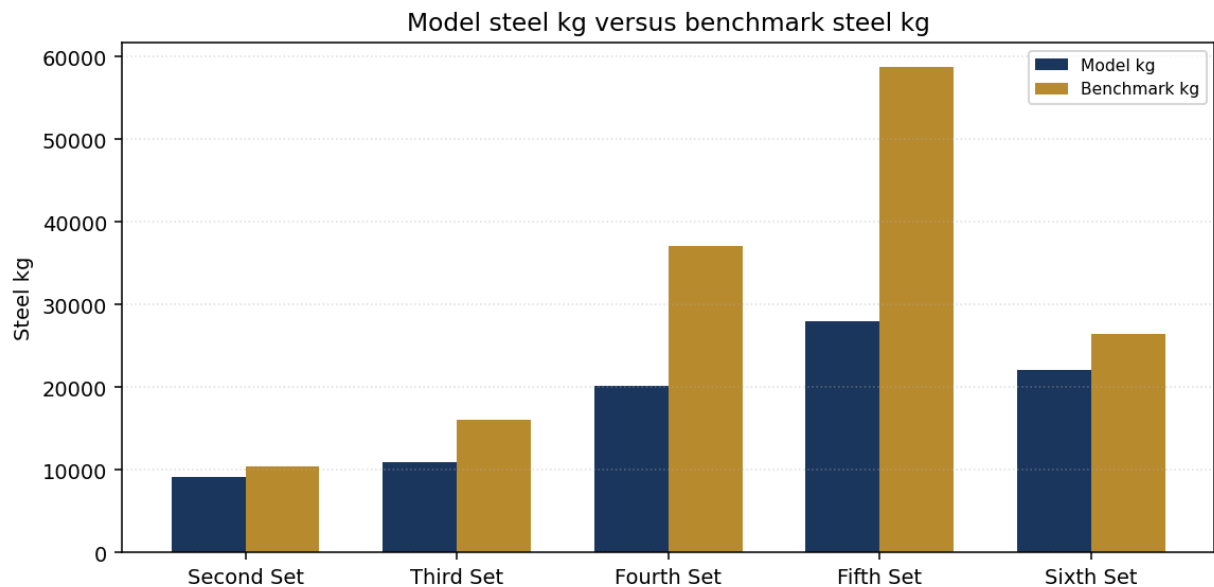
Drawing set	Beam ID	Bar ID	Correct of detected	Diameter ID	Steel	Overall
Second Set	95.52%	65.28%	31.35%	78.97%	87.95%	70.03%
Third Set	90.48%	54.29%	27.27%	74.31%	68.25%	60.07%
Fourth Set	78.32%	40.43%	41.32%	75.26%	54.29%	53.59%
Fifth Set	76.47%	39.29%	36.89%	76.75%	47.58%	50.06%
Sixth Set	95.86%	59.83%	39.38%	81.85%	83.18%	69.56%



6. Pooled Second-to-Sixth performance

KPI	Pooled result	Raw counts
Beam identification	85.12%	515.0 / 605.0
Bar identification	48.17%	2008.0 / 4169.0
Correct of detected bars	36.50%	733.0 / 2008.0
Diameter identification	77.84%	1563.0 / 2008.0
Steel / weight accuracy	60.64%	90305.121 / 148910.644 kg
Overall accuracy	57.61%	Mean of four pooled KPIs; diameter excluded

Headline pooled results sum raw numerators and denominators. Steel accuracy is computed after pooling kg. Set percentages are not averaged for the headline overall score.



7. Diameter identification (detected bar lines)

QA.2A's published field diameter_accuracy_pct is an alias of bar matching accuracy (MATCH / detected) and is not used here. Diameter identification below is counted from Bar Matching rows: a detected bar is diameter-correct unless status is WRONG_DIAMETER. GT diameter is the estimator line diameter. Percentages are pooled from raw counts across Second–Sixth; set percentages are not averaged. Diameter remains excluded from overall.

Diameter	GT bar lines	Detected	MATCH	WRONG_DIA	Diameter ID	Note
Ø8	358	152	32	3	98.03%	Usually right when found; many missing
Ø10	714	214	56	11	94.86%	Usually right when found; many missing
Ø12	573	207	68	28	86.47%	Usually right when found; many missing
Ø16	588	372	169	112	69.89%	Frequent diameter swaps
Ø20	784	436	193	155	64.45%	Frequent diameter swaps
Ø25	1,097	616	210	130	78.90%	
Ø32	26	11	5	6	45.45%	Low volume - percentage unstable
TOTAL scored Ø8-Ø32	4,140	2,008	733	445	77.84%	Pooled headline from raw counts

8. Diameter-wise steel quantity — not the same as diameter identification

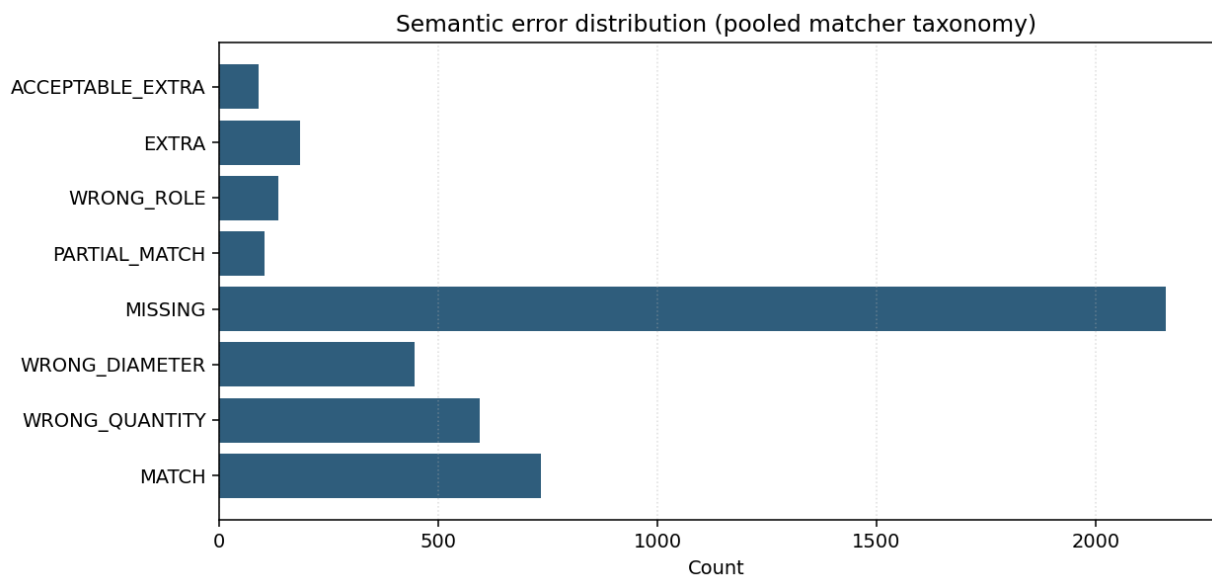
Quantity ratio = automated kg / estimated kg × 100. It is not accuracy. A ratio above 100% is an overestimate. This table pools kilogram totals by diameter before computing difference and ratio. It is not diameter identification.

Diameter	Estimated kg	Automated kg	Difference kg	Abs % diff	Quantity ratio
Ø8	6,515	5,051	–1,464	22.5%	78%
Ø10	24,555	14,163	–10,393	42.3%	58%
Ø12	22,253	12,722	–9,531	42.8%	57%
Ø16	11,598	7,678	–3,920	33.8%	66%
Ø20	35,888	20,506	–15,382	42.9%	57%
Ø25	45,294	28,282	–17,011	37.6%	62%
Ø32	2,808	1,901	–906	32.3%	68%
TOTAL	148,911	90,303	–58,607	39.4%	61%

9. Semantic error profile

Matcher taxonomy	Pooled count
MISSING	2161
MATCH	733
WRONG_QUANTITY	593
WRONG_DIAMETER	445
EXTRA	183
WRONG_ROLE	134
PARTIAL_MATCH	103
ACCEPTABLE_EXTRA	89

MISSING bars are ground-truth lines the model did not identify. MATCH / WRONG_QUANTITY / WRONG_DIAMETER / WRONG_ROLE / PARTIAL_MATCH describe detected bars. EXTRA is a model line without a GT pair. Deterministic disagreement is not treated as proof that Vision is wrong.



10. Engineering calculation profile

Engineering condition	Count
CUT_LENGTH_DERIVED_FALLBACK	188
CUT_LENGTH_UNAVAILABLE	0
STIRRUP_ENGINEERING_UNAVAILABLE	69
PARTIAL_CALCULATION	82
INCOMPATIBLE	0
SPACER_ZERO	242

Spacers remain deterministic-only. Stirrup identification may come from Vision; stirrup geometry, legs, zones, spacing, quantity, hooks, cut length and weight remain deterministic. Vision never invents stirrup engineering quantities. Semantic/engineering stirrup conflicts are recorded, not silently overwritten.

11. HYBRID / FALLBACK coverage

HYBRID_ONLY scores use the valid matched benchmark subset for Vision-usable beams. FALLBACK_ONLY uses beams that applied the D.1 deterministic fallback. Full-population scores are not replaced by HYBRID_ONLY steel accuracy. A zero cohort is reported as NOT APPLICABLE.

Cohort	Beam ID	Bar ID	Correct	Diameter	Steel	Overall
HYBRID_ONLY	97.52%	57.03%	36.39%	78.46%	75.22%	66.54%

FALLBACK_ONLY	97.73%	48.43%	38.21%	68.29%	71.00%	63.84%
FULL_POPULATION	85.12%	48.17%	36.50%	77.84%	60.64%	57.61%

12. Current workflow value

Positioning: model-assisted estimation. Estimator verification is required. No current time-and-motion study exists in this phase, so accuracy is not equivalent to time saved. Vision API success is not engineering accuracy. Fallback coverage is part of the measured system, not hidden Vision performance.

13. Methodology, sources and limitations

P2.6.10-E.3 uses QA.2A BeamMatcher / BarMatcher / MetricsEngine metric8 and the QA.3.0 four-KPI overall mean. Ground truth is evaluation-only (estimator Excel; Second–Third also have a validated QA.2B.1 context path where discovered). Truth is never used in runtime semantic resolution. Limitations: ESTIMATOR_WORKBOOK_MAPPING_LIMITATION, FIRST_SET_EXCLUDED, SHADOW_ONLY_NOT_PRODUCTION, FALLBACK_POPULATION_PRESENT. PRODUCTION_WRITE = false. ENGINEERING_CHANGES = NONE. Shadow-only; not production routing.